DYNAMIC ARP INSPECTION (DAI)

Exam-Focused Computer Networks Notes

Short explanations • Cisco IOS commands • Attack flow • Original lecture screenshots

Quick Revision Map

- ARP basics: map a known IPv4 address to a MAC address on the local LAN.
- Gratuitous ARP (GARP): an unsolicited ARP announcement/reply; useful but can be abused.
- ARP poisoning: fake ARP information redirects traffic through an attacker.
- DAI: validates ARP packets arriving on untrusted switch ports.
- DHCP Snooping binding table: normally supplies IP–MAC–VLAN–interface information for DAI.
- Trusted ports: DAI does not inspect ARP packets arriving on these ports; configure carefully.
- Static-IP devices: may need an ARP ACL because they might not appear in the DHCP Snooping table.
- Rate limiting and optional validation checks add protection against floods and malformed/inconsistent ARP packets.

1. ARP Review — Why ARP Is Needed

Concept

Address Resolution Protocol (ARP) lets an IPv4 device discover the MAC address associated with a known IPv4 address on the same local Layer 2 network. IP addresses identify the Layer 3 destination; Ethernet frames still need a destination MAC address for the next local hop.

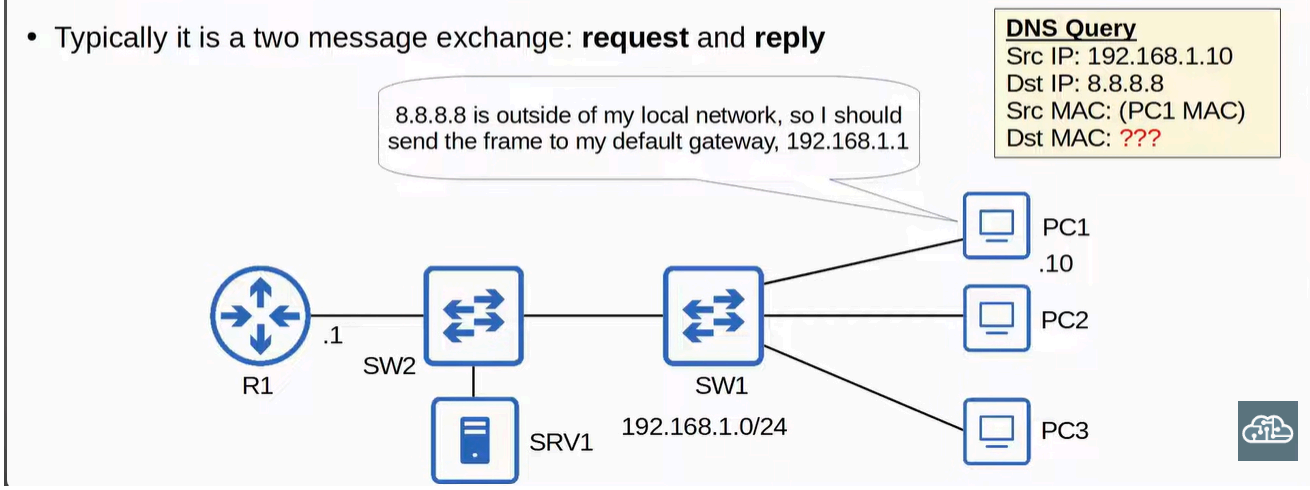
Example: PC1 wants to reach 8.8.8.8

- PC1 = 192.168.1.10/24; default gateway = 192.168.1.1.
- Because 8.8.8.8 is outside 192.168.1.0/24, PC1 sends the packet to its default gateway.
- PC1 first needs the gateway's MAC address, so it broadcasts an ARP Request: “Who has 192.168.1.1?”
- R1 replies with its MAC address. PC1 stores the IP-to-MAC mapping in its ARP cache.
- The DNS query's IP destination remains 8.8.8.8, but the Ethernet destination MAC is R1's MAC for this first hop.

Key distinction: ARP resolves the MAC address of the next hop, not the MAC address of a remote Internet server.

ARP Review

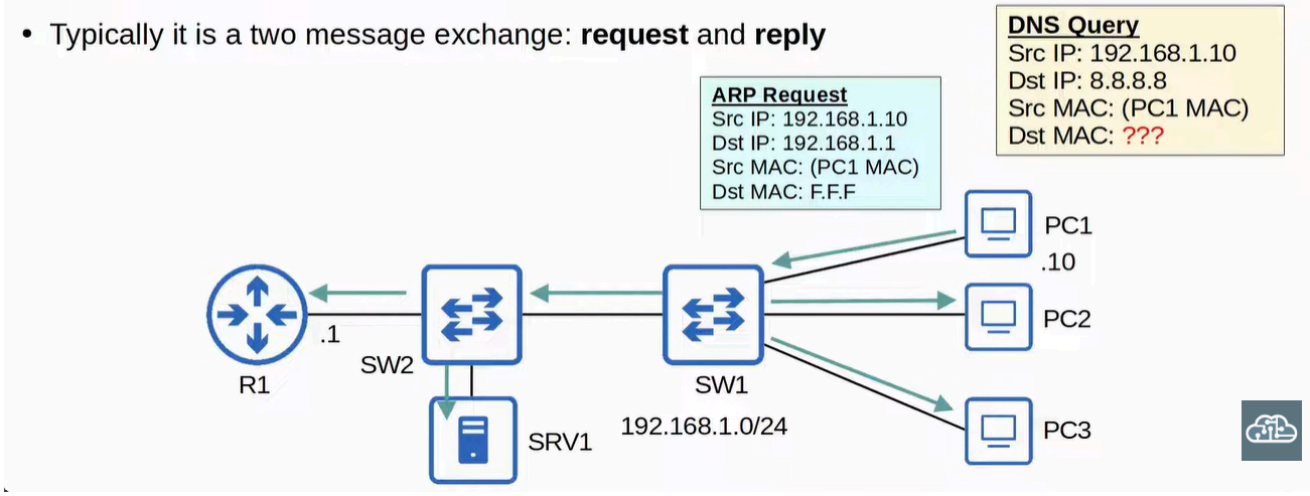
- ARP is used to learn the MAC address of another device with a known IP address.
- For example, a PC will use ARP to learn the MAC address of its default gateway to communicate with external networks.
- Typically it is a two message exchange: **request** and **reply**



Original image 1 — ARP review topology and the default-gateway decision.

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Original image 2 — ARP Request broadcast through the local switched network.

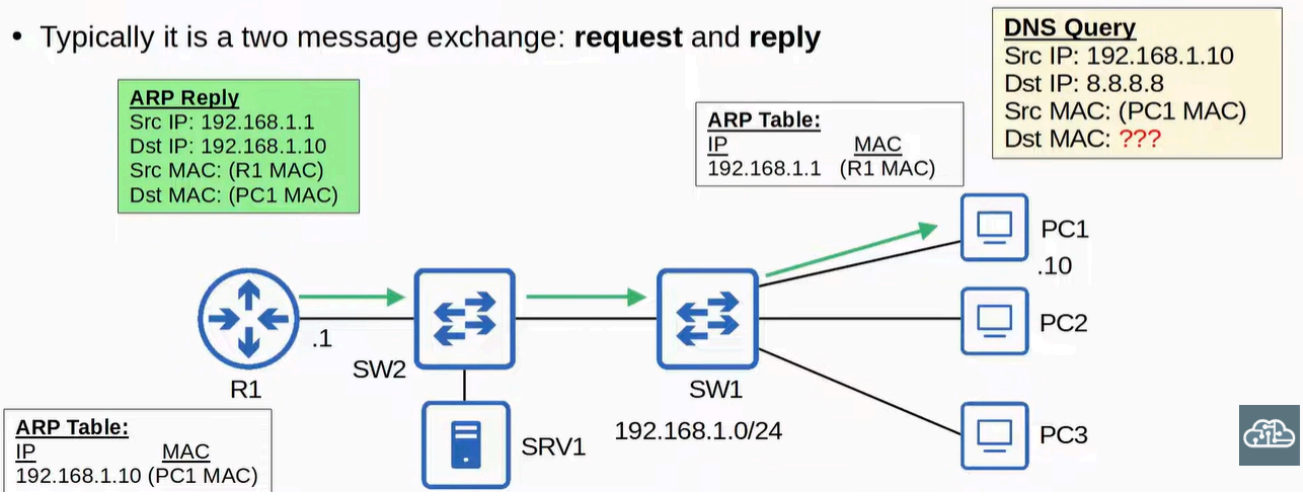
ARP Review

```
> Frame 99: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
v Ethernet II, Src: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00)
    Type: ARP (0x0806)
    Padding: 000000000000000000000000000000000000
v Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00)
    Sender IP address: 192.168.1.10
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.1
```

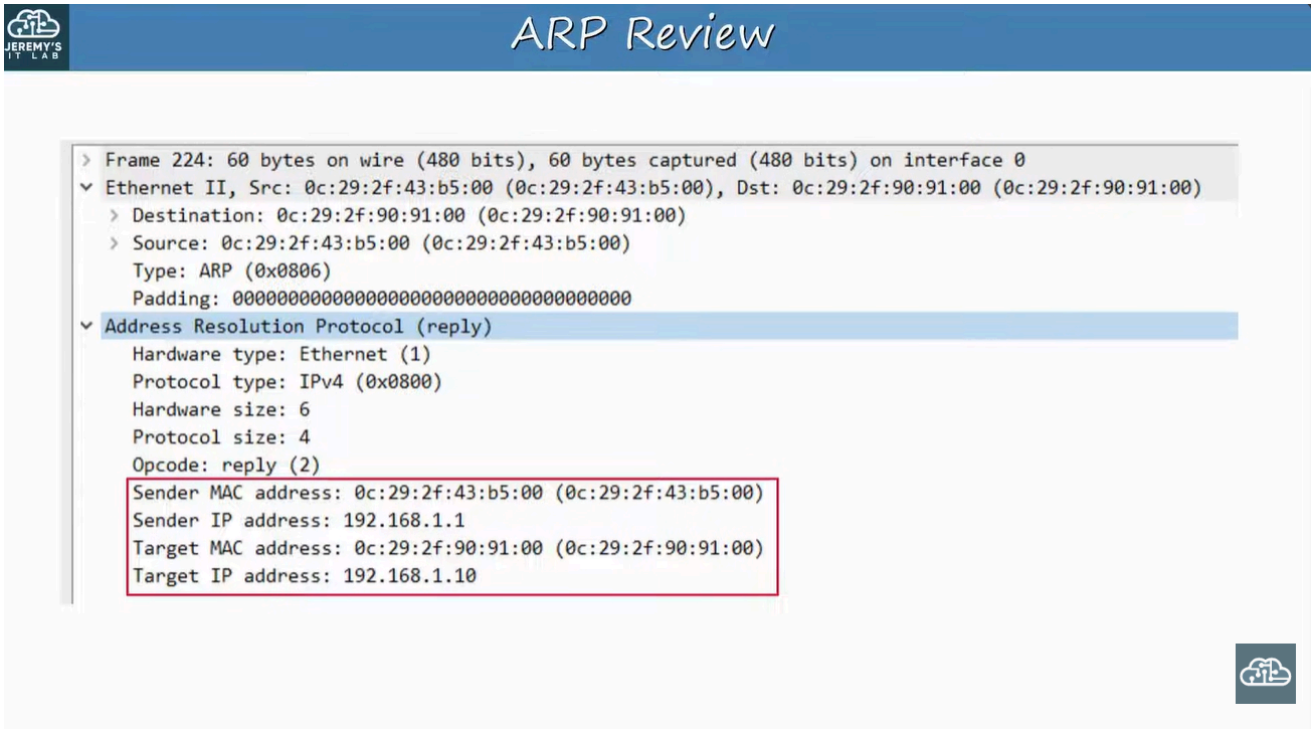
Original image 3 — Wireshark view of an ARP Request: sender and target fields.

ARP Review

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- For example, a PC will use ARP to learn the MAC address of its default gateway to communicate with external networks.
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Original image 4 — ARP Reply returns the gateway's MAC address.



Original image 5 — Wireshark view of an ARP Reply.

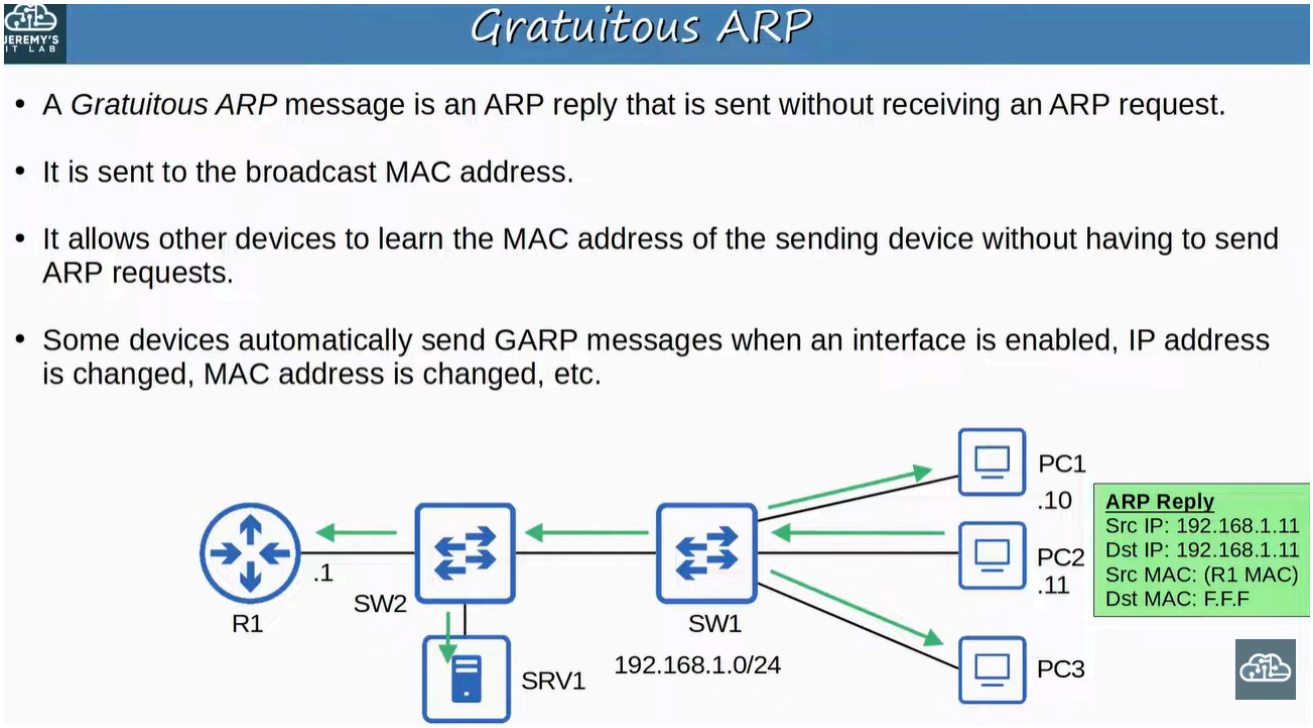
2. Gratuitous ARP (GARP)

Definition

A Gratuitous ARP is an ARP announcement sent without a preceding ARP request. It is commonly broadcast so other devices can learn or refresh the sender's IP-to-MAC mapping.

Key points

- Can help announce an IP/MAC mapping when an interface comes up or an address changes.
- Can help detect duplicate IPv4 addresses in some implementations.
- Because devices may update their ARP caches based on such messages, attackers can abuse forged GARP messages for ARP poisoning.
- GARP is not automatically malicious; the context and validity of the message matter.



Original image 6 — Gratuitous ARP: an unsolicited ARP announcement.

3. ARP Poisoning / Man-in-the-Middle

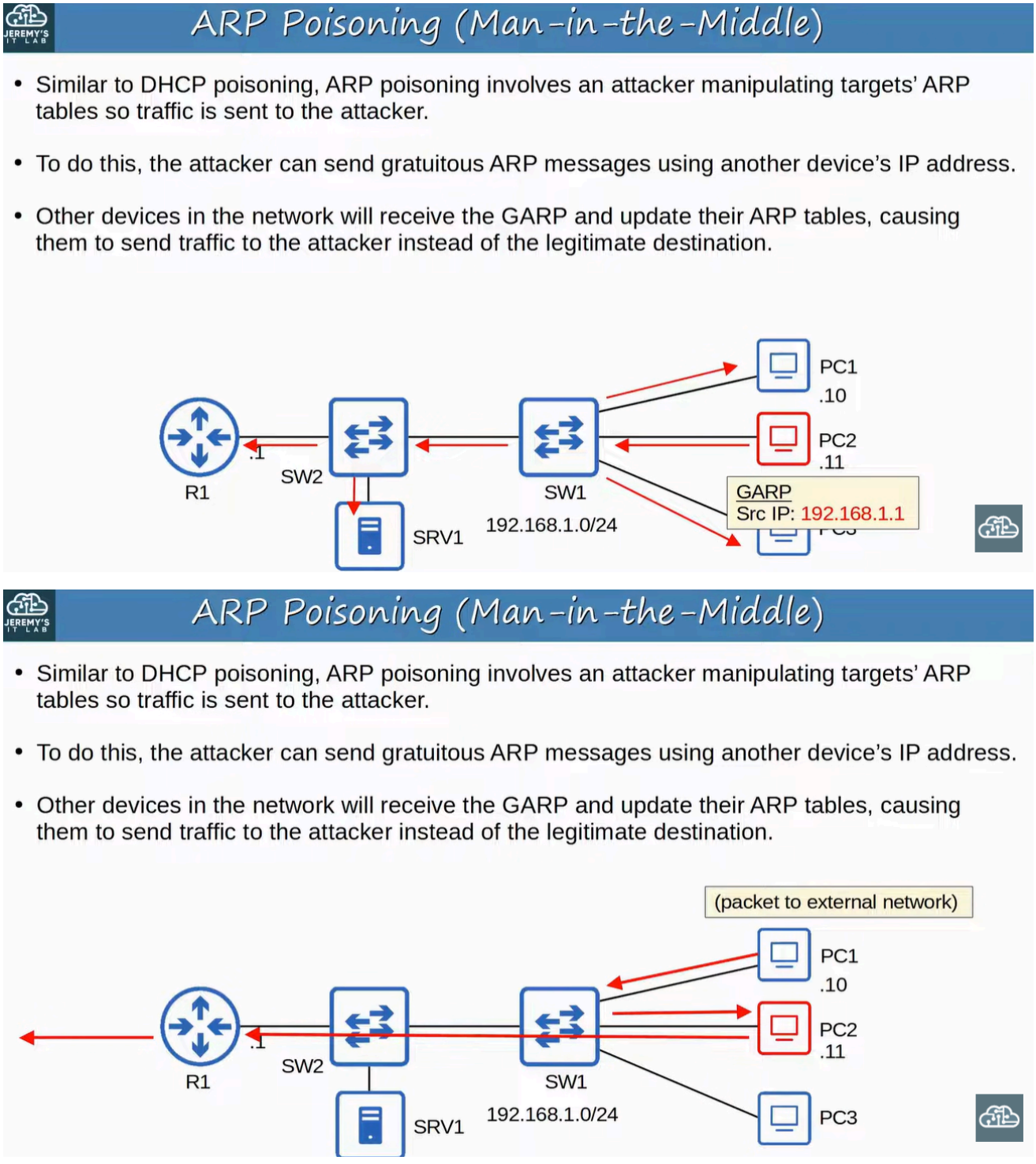
Concept

ARP poisoning (ARP spoofing) occurs when an attacker sends false ARP information so that a victim associates a legitimate IP address—often the default gateway—with the attacker's MAC address. The victim may then send traffic to the attacker.

Example attack flow

- Legitimate gateway: 192.168.1.1 → R1's real MAC.
- Attacker PC2 sends forged ARP messages claiming that 192.168.1.1 is at PC2's MAC.
- Victim PC1 may update its ARP cache with the false mapping.
- Traffic intended for the gateway can be redirected to PC2, enabling interception or disruption.

Exam keyword: Man-in-the-Middle (MITM). ARP itself has no built-in authentication, so switches can use DAI to validate ARP traffic.



Original image 8 — Example of redirected traffic to an external network.

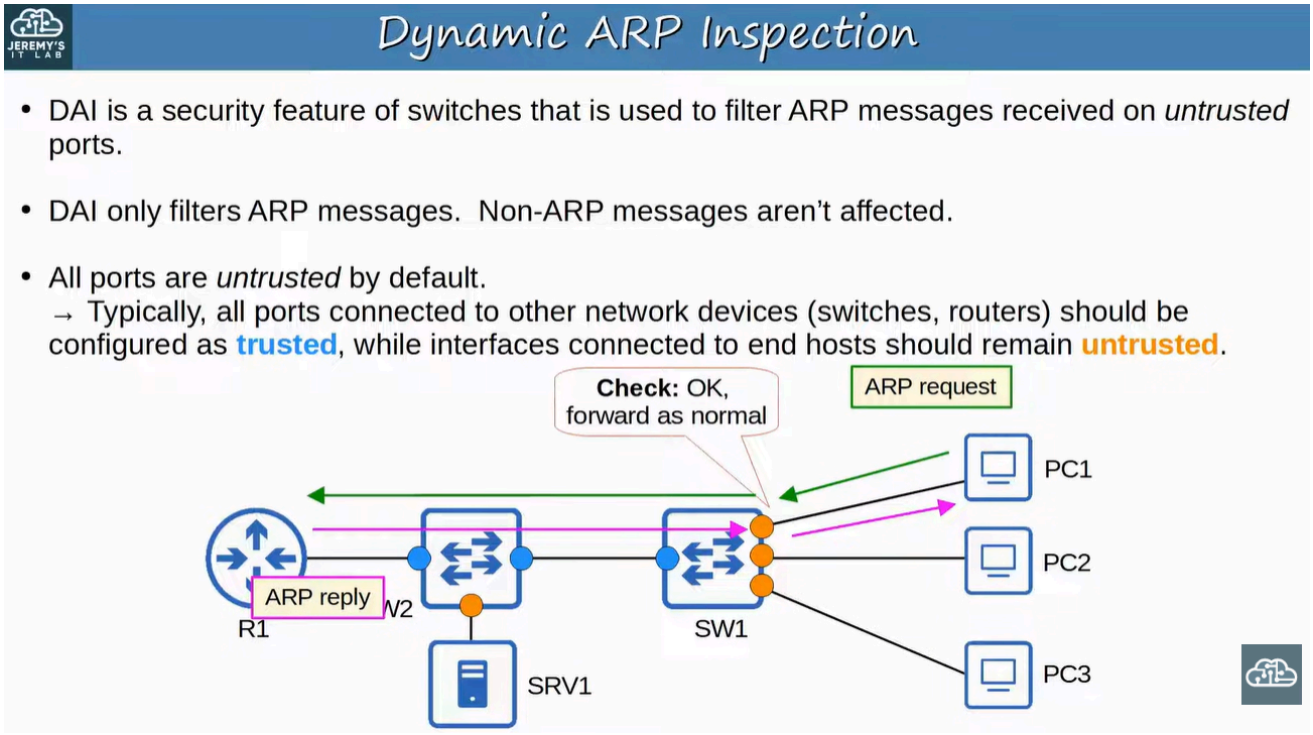
4. Dynamic ARP Inspection (DAI)

Definition

Dynamic ARP Inspection is a Layer 2 switch security feature that checks ARP messages received on untrusted ports and drops invalid messages to help prevent ARP spoofing.

How it works

- DAI examines ARP messages received on untrusted interfaces.
- By default, it compares the ARP sender IP and sender MAC information with the DHCP Snooping binding database.
- A matching entry is normally forwarded; a mismatch or missing valid binding is normally dropped, subject to platform/configuration and any configured ARP ACL.
- ARP packets arriving on trusted interfaces bypass DAI inspection and are forwarded normally.
- Non-ARP traffic is not filtered by DAI.



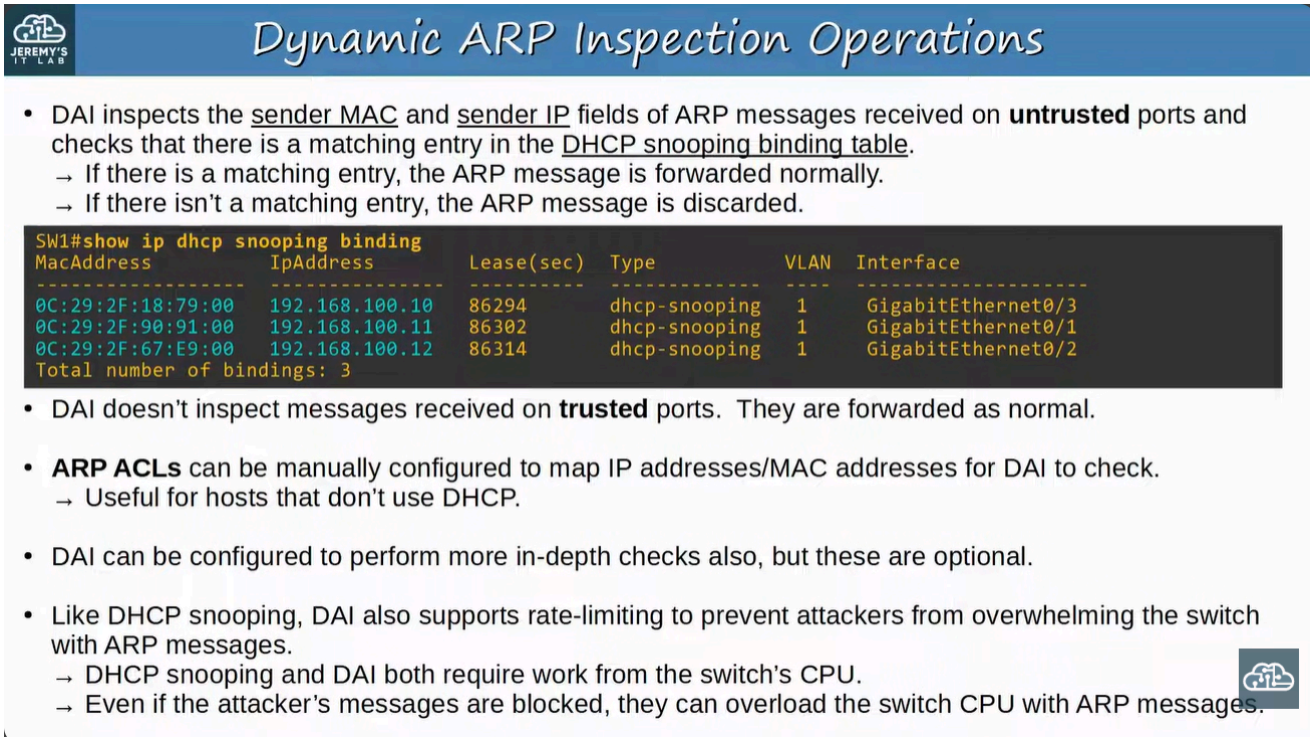
Original image 9 — DAI checks ARP packets on untrusted ports; trusted ports bypass inspection.

Original image 10 — DAI operation and example DHCP Snooping binding entries.

5. Trusted vs Untrusted Ports

Port type	Typical connection	DAI behavior
Untrusted (default)	End hosts such as PCs	ARP packets are inspected.
Trusted (explicitly configured)	Uplink toward a router or another switch, when appropriate	ARP packets are not inspected by DAI on ingress.

Important: "trusted" is a security decision, not a label to apply to every network-device port automatically. Trust only links where bypassing inspection is justified.



Original image 11 — Cisco DAI configuration example for VLAN 1 and trusted uplinks.

6. Cisco Configuration — Enable DAI

DAI is enabled per VLAN. The screenshot's example uses VLAN 1; replace 1 with the VLAN ID used in your lab.

Enable ARP inspection

```
Switch(config)# ip arp inspection vlan 1
```

Enables DAI for VLAN 1. Configure globally on the switch.

Trust an appropriate uplink

```
Switch(config)# interface g0/0
Switch(config-if)# ip arp inspection trust
```

Marks this interface as trusted for DAI. Interface names vary by switch model.

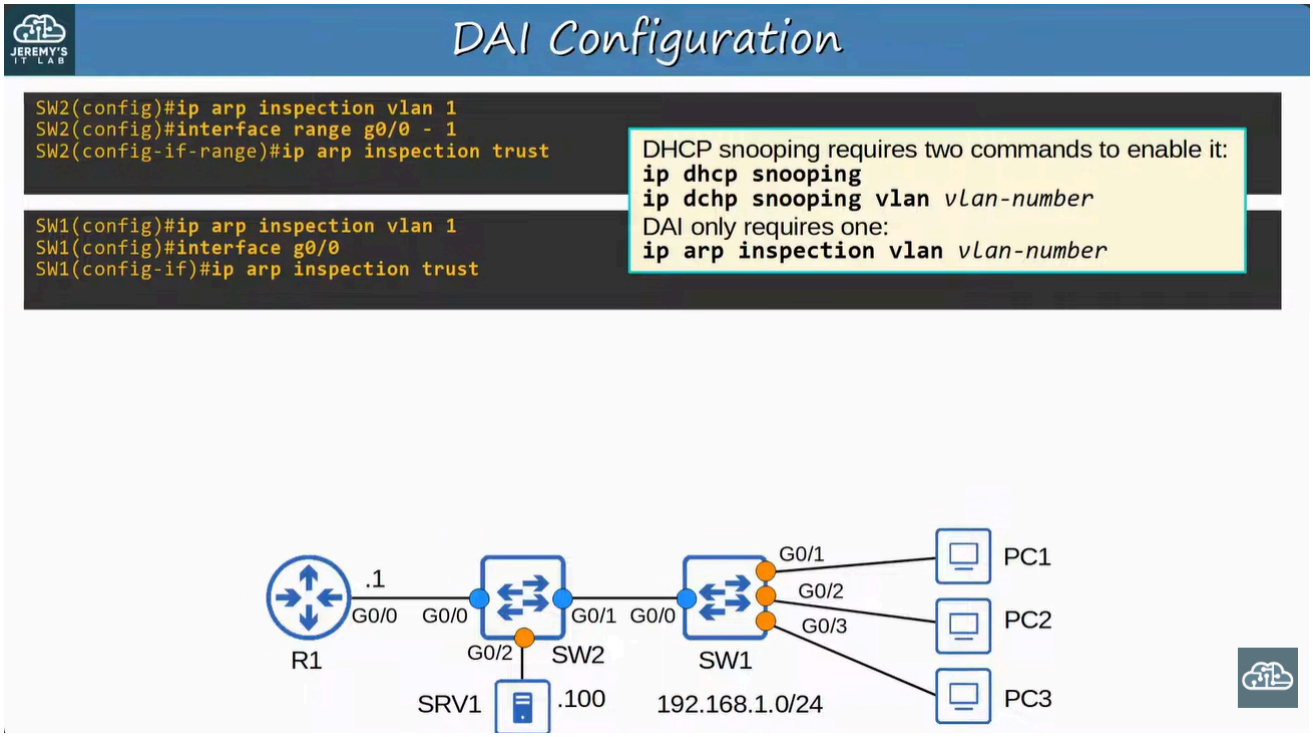
The screenshot also shows a range command. Use interface-range syntax only when all selected ports should have the same trust state:

```
Switch(config)# interface range g0/0 - 1
Switch(config-if-range)# ip arp inspection trust
```

DHCP Snooping is usually needed for the binding database used by DAI. Enable it for the relevant VLAN as well:

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 1
```

On a DHCP server-facing port, configure DHCP Snooping trust according to the topology; DHCP Snooping trust and DAI trust are separate settings.



Original image 12 — DAI configuration commands and the DHCP Snooping enablement reminder.

7. Verify DAI and Rate Limiting

Useful verification commands

```
Switch# show ip arp inspection
Switch# show ip arp inspection interfaces
Switch# show ip dhcp snooping binding
Switch# show running-config | include validate
Switch# show errdisable recovery
```

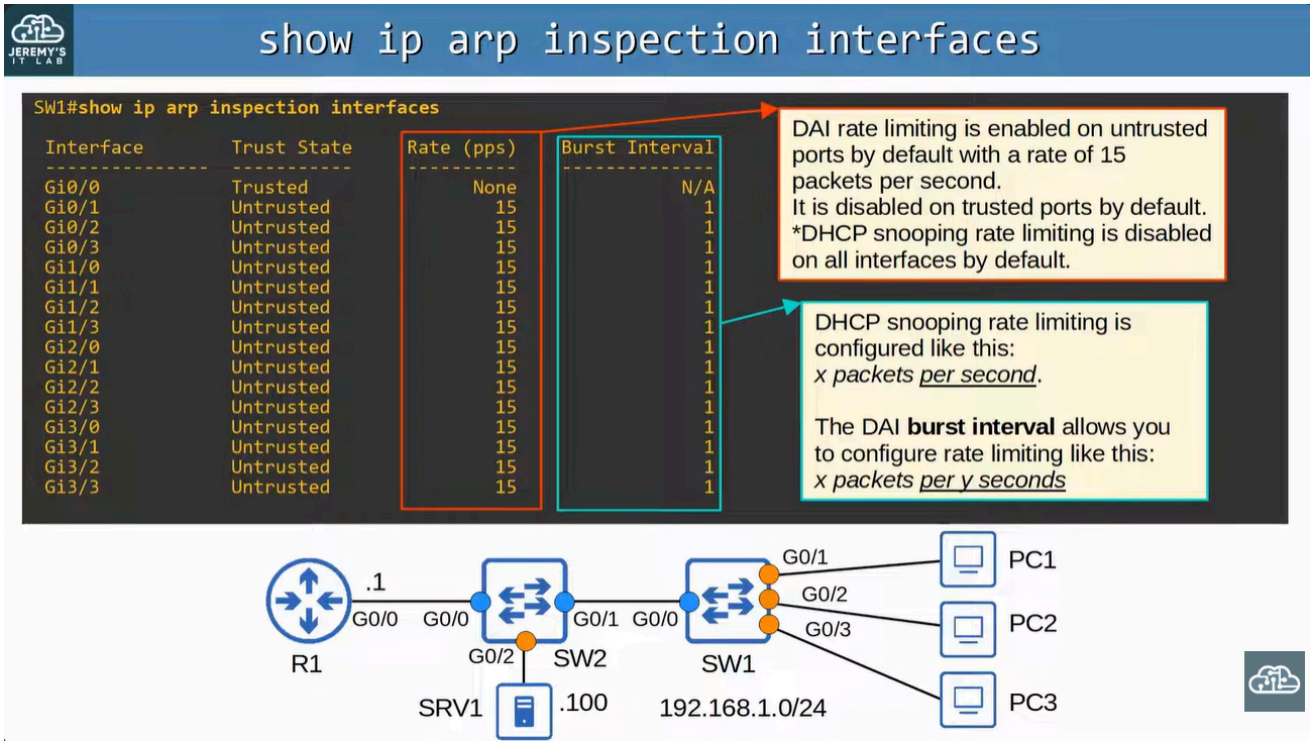
DAI rate limiting

Rate limiting restricts how many ARP packets an interface can receive in a specified interval. It helps protect switch CPU resources from excessive ARP traffic. The screenshot shows a common default of 15 packets per second on untrusted ports, but defaults can vary by platform and software.

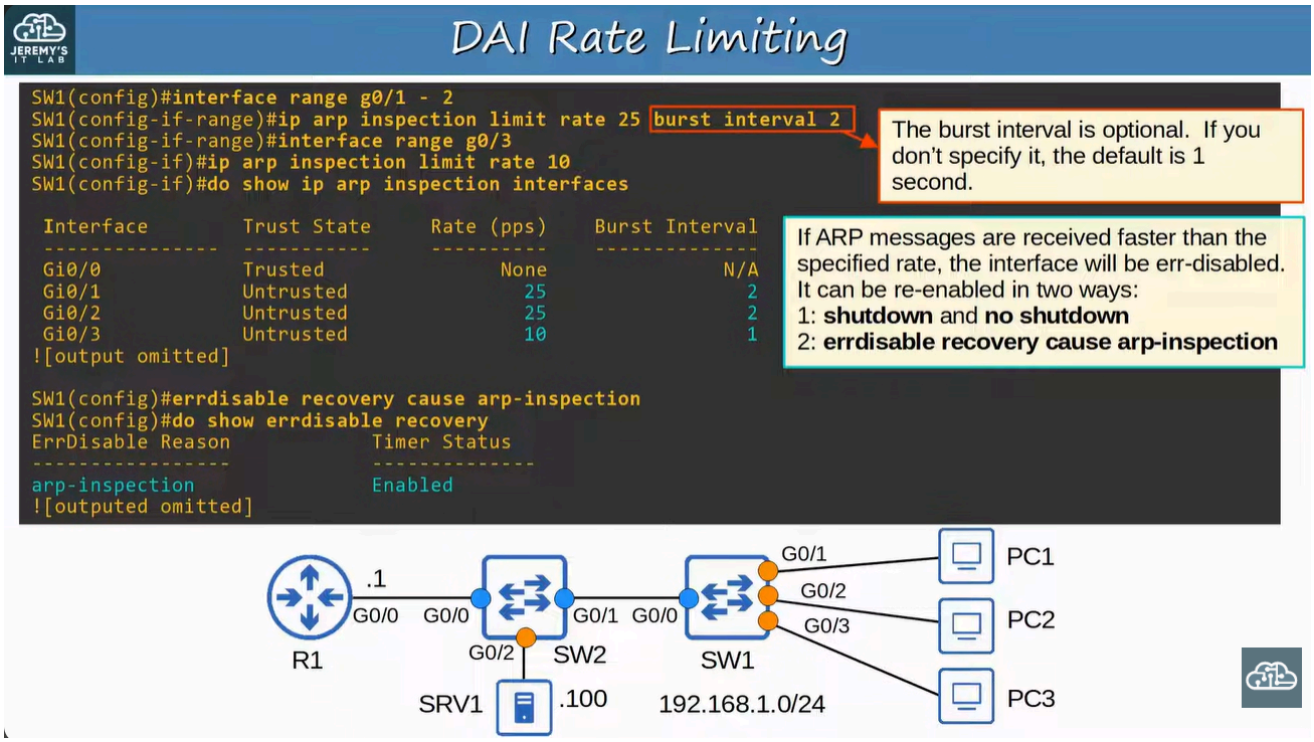
```
Switch(config)# interface range g0/1 - 2
Switch(config-if-range)# ip arp inspection limit rate 25 burst interval 2
Switch(config)# interface range g0/3
Switch(config-if-range)# ip arp inspection limit rate 10
```

If the configured rate is exceeded, the switch may place the interface into an error-disabled state. Recovery can be manual (shutdown/no shutdown) or configured, depending on platform and policy:

```
Switch(config)# errdisable recovery cause arp-inspection
Switch# show errdisable recovery
```



Original image 13 — DAI interface trust states and rate-limit defaults shown in command output.



Original image 14 — Configuring DAI rate limits and error-disabled recovery.

8. Optional DAI Validation Checks

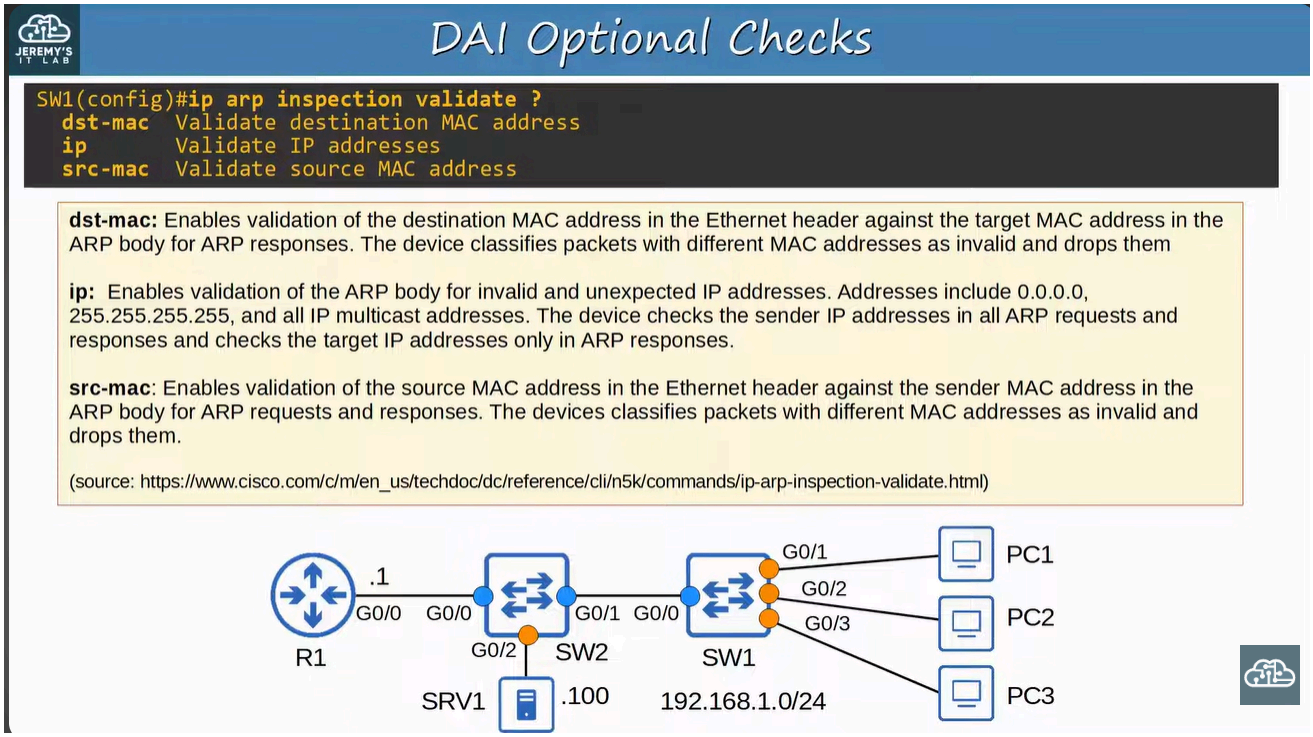
Besides the standard sender IP/MAC check, Cisco IOS can be configured to perform additional consistency checks. These checks are optional and must be configured together in one command; specifying a later command replaces the configured list rather than simply appending one option.

Option	What it checks	Exam memory cue

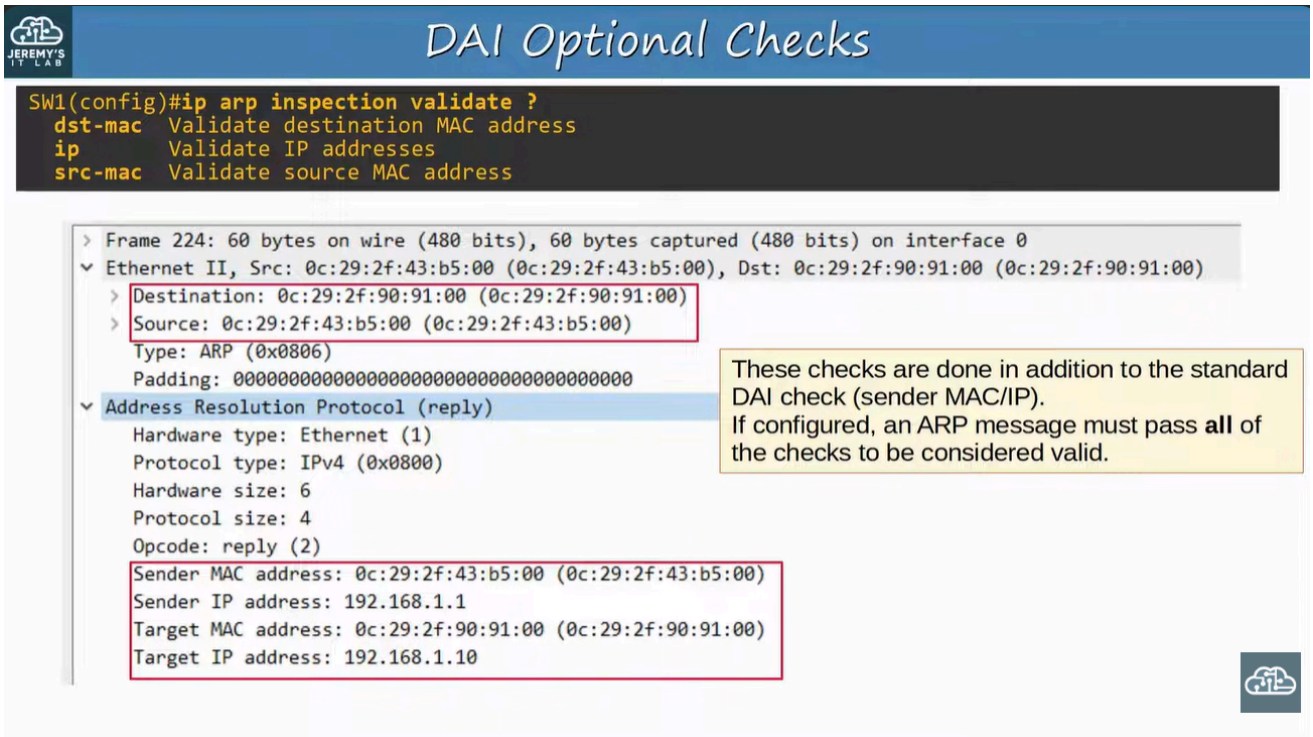
src-mac	Ethernet source MAC matches the ARP sender MAC.	Source MAC consistency
dst-mac	For ARP replies, Ethernet destination MAC matches the ARP target MAC.	Destination MAC consistency
ip	Checks for invalid/unexpected IP addresses in ARP fields.	IP field validity

```
Switch(config)# ip arp inspection validate src-mac dst-mac ip
Switch# show running-config | include validate
```

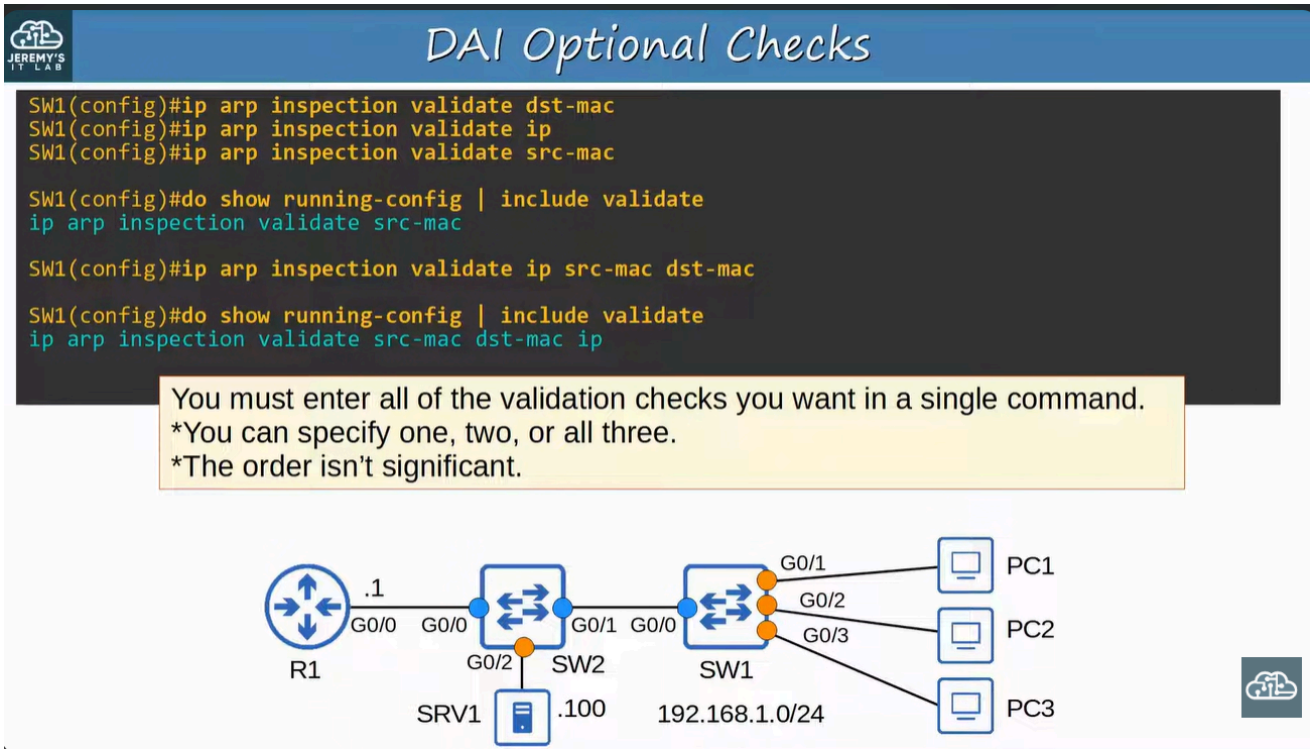
All configured validation checks must pass for the packet to be considered valid.



Original image 15 — Optional DAI validation options and their meanings.



Original image 16 — Example packet fields used for validation.



Original image 17 — Configuring one, two, or all three validation checks.

9. ARP ACLs — Supporting Devices with Static IPs

Why use an ARP ACL?

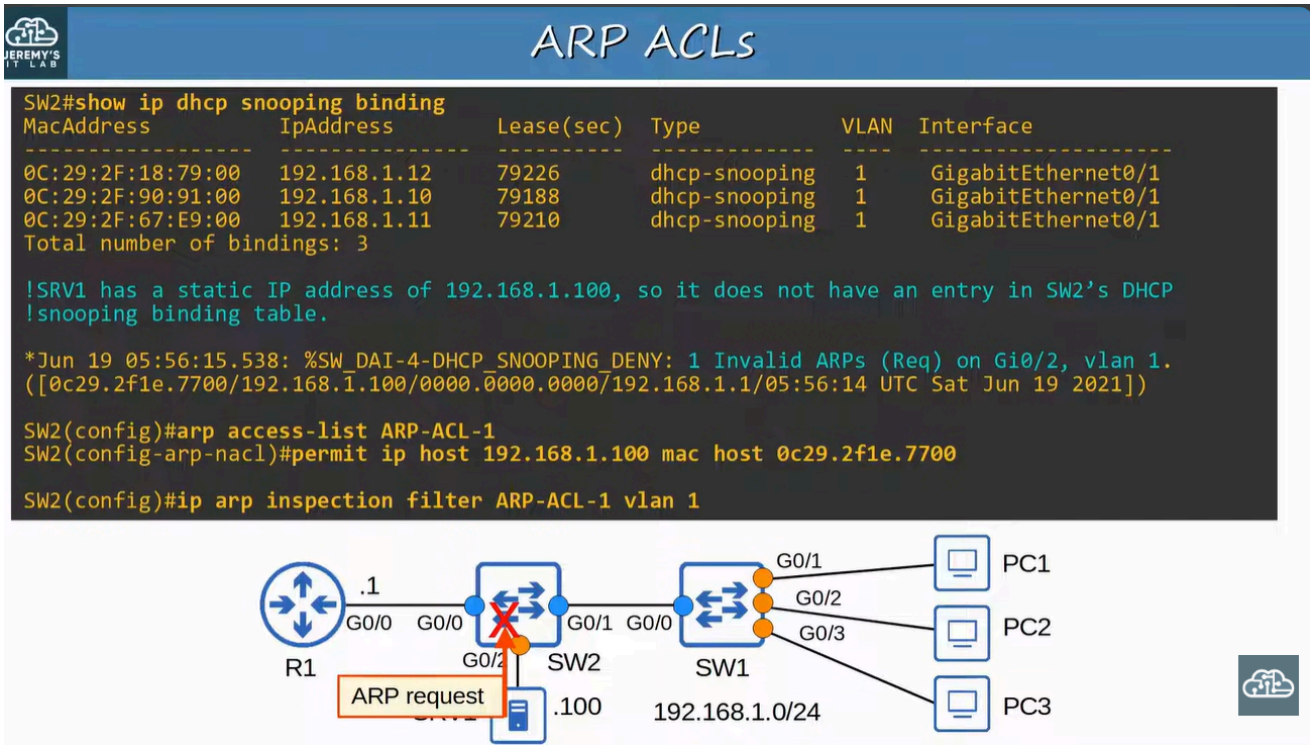
A device configured with a static IP may not appear in the DHCP Snooping binding table. If DAI cannot validate its ARP message against a binding, the message may be dropped. An ARP ACL can explicitly permit a known IP/MAC pair.

Example from the screenshot

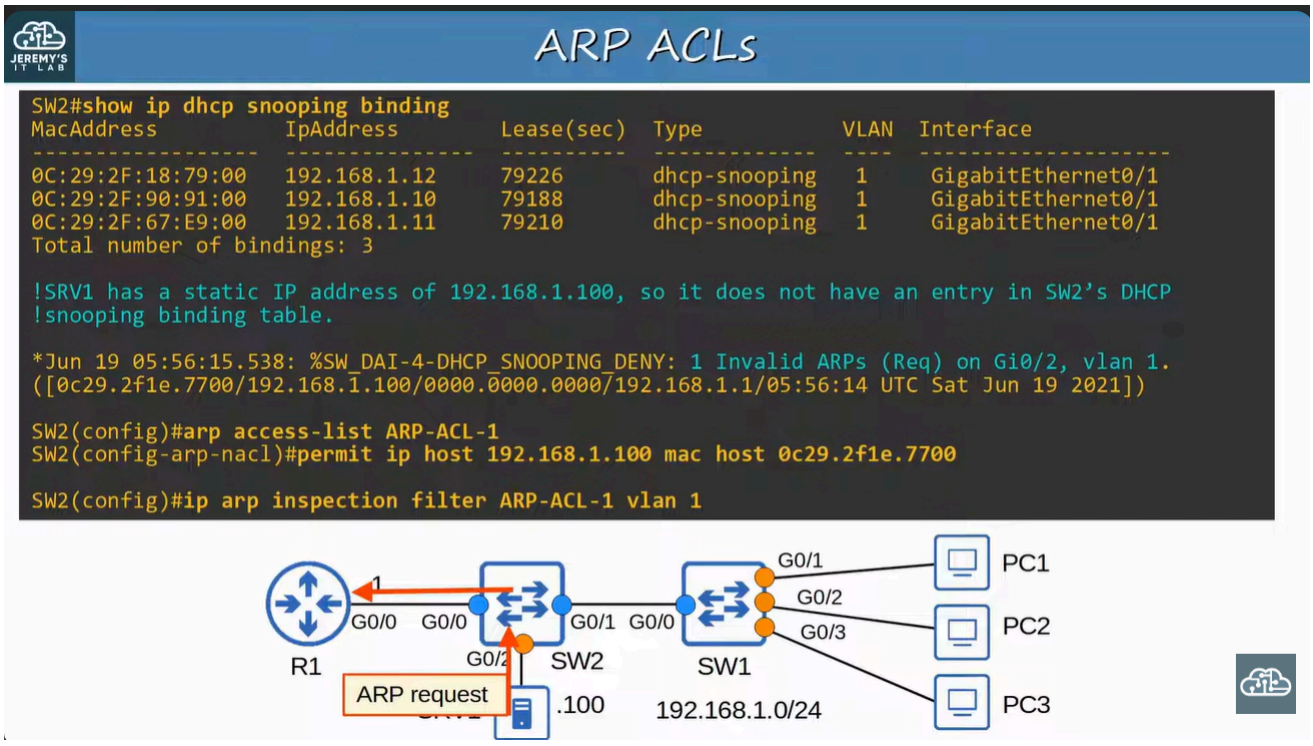
```
Switch(config)# arp access-list ARP-ACL-1
Switch(config-arp-nacl)# permit ip host 192.168.1.100 mac host 0c29.2f1e.7700
Switch(config)# ip arp inspection filter ARP-ACL-1 vlan 1
```

Meaning: create an ARP ACL named ARP-ACL-1, permit the specified IP/MAC pair, and apply the ACL to VLAN 1. Verify that the MAC address and syntax match the actual device and IOS version.

Static ACL behavior matters: depending on the configured static-ACL option, an ARP ACL can be used as the authoritative permit list, so entries not permitted by it may be denied rather than falling back to the DHCP Snooping table. Check the platform's command reference before deploying.



Original image 18 — ARP ACL example for a statically addressed server.



Original image 19 — DAI status/counters showing forwarded, dropped, and ACL-permitted ARP packets.

10. Exam Comparison and Final Revision

Feature	Main job	Remember
ARP	Find a MAC address for a known local IPv4 next hop.	Request + reply
Gratuitous ARP	Announce/refresh an IP-to-MAC mapping without a request.	Unsolicited ARP

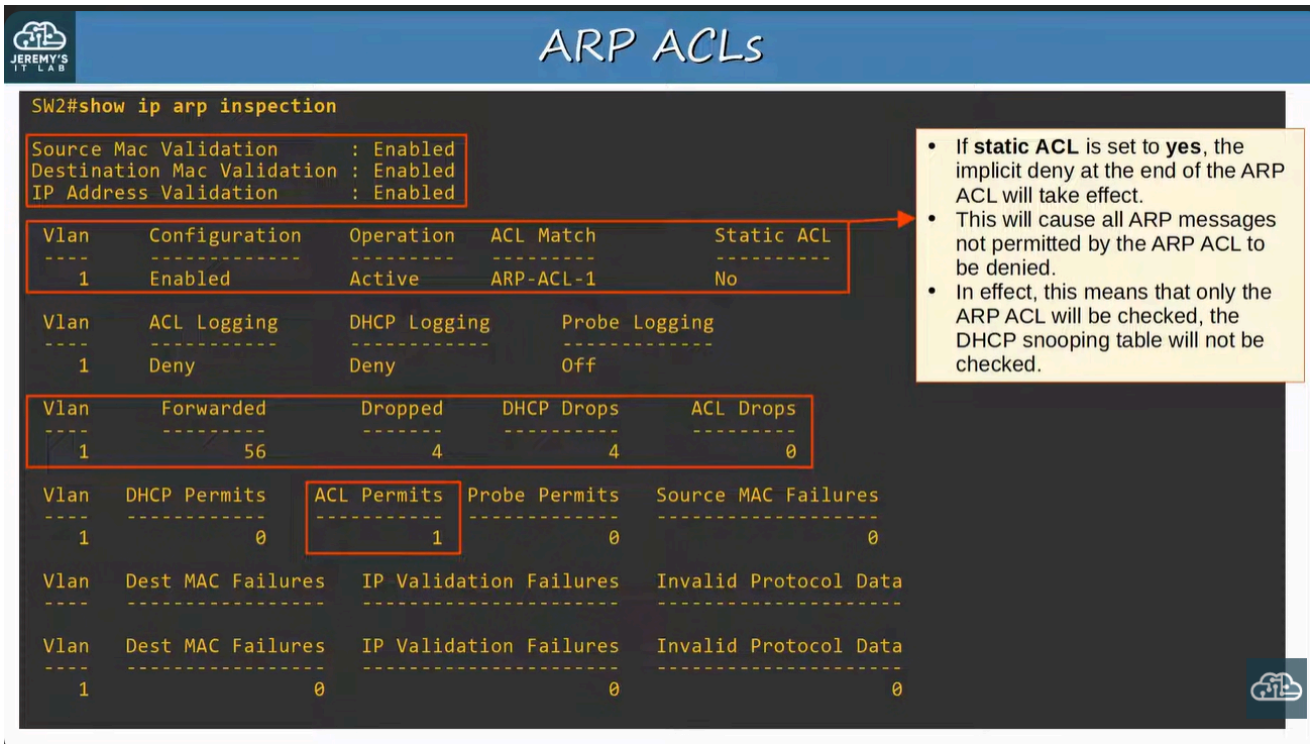
ARP poisoning	Falsify ARP mappings to redirect or disrupt traffic.	MITM risk
DHCP Snooping	Builds a binding database from DHCP exchanges.	IP + MAC + VLAN + port
DAI	Validates ARP messages on untrusted ports.	Match binding or ARP ACL
ARP ACL	Provides explicit IP/MAC permissions, including static-IP hosts.	Manual mapping

Likely exam questions

- What is DAI, and which attack does it help prevent?
- Which ports are trusted and untrusted by default?
- What information does DAI compare against the DHCP Snooping binding database?
- Why might a statically addressed server need an ARP ACL?
- What is the difference between DHCP Snooping and DAI?
- What is the purpose of DAI rate limiting?
- What do src-mac, dst-mac, and ip validation check?

One-minute memory trick

DHCP Snooping = builds the IP/MAC/VLAN/port bindings. DAI = checks ARP claims. ARP ACL = manual exception for known mappings. Rate limiting = controls ARP packet bursts. Trusted port = bypasses DAI inspection on ingress.



Original image 20 — Final ARP ACL example and configuration context.